

Explicit Viability Bounds for Safety Filters with Configuration-Dependent Braking Authority

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Abstract—A cable-suspended payload may approach a wall while every cable pulls it toward the wall; the quadrotors must first swing behind the payload before they can brake. A high-order control barrier function (HOCBF) filter that admits an input now can lose feasibility during this swing, and we give a condition under which it must, with a two-quadrotor example. We then derive an explicit certificate. A thrust-reserved swing-and-brake maneuver that carries the weight of the payload has an exactly computable stopping distance, which yields an authority barrier and two altitude barriers; their certified set is controlled invariant for the continuous-time taut-cable model, provided that closed-loop solutions exist, and the braking maneuver is always feasible for the resulting filter. An optimistic stopping distance bounds the viability kernel from outside. Both distances use the support function of the zonotope of tension forces, which depends on the cable directions. On the exact model with a hold of 1 ms, one of 100 trials started at the boundary of the certified set reaches the wall, by 2 μ m. Simulations expose substantial conservatism when swinging is required, and wall contacts under a slower sampled filter and a slack-cable model.

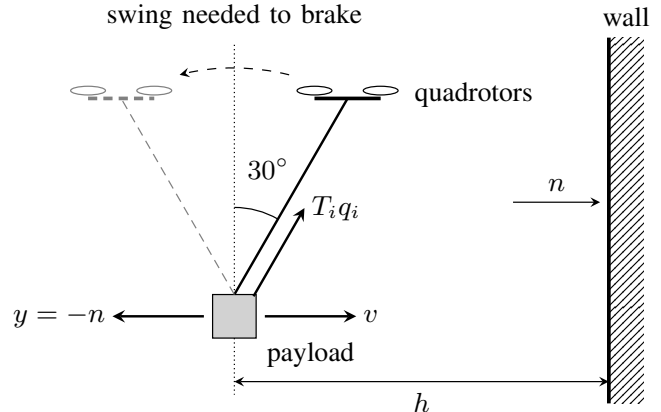


Fig. 1. Two quadrotors carry a payload toward a wall. With both cables leaning toward the wall, positive tensions cannot brake it; a cable must first swing to the far side (dashed).

I. INTRODUCTION

A quadrotor can pull a cable-suspended payload but cannot push it. If all quadrotors are between a moving payload and a wall, changing the tensions cannot brake the payload; a quadrotor first has to swing to its far side (Fig. 1). This delay is hidden by a high-order control barrier function (HOCBF) that tests only the currently achievable acceleration. We quantify the delay and construct a safety certificate from the maximum wallward excursion of a feasible braking maneuver.

Input-constrained HOCBFs [1], [2], feasibility constraints [3], and backup, constructive, and predictive CBFs [4]–[7] address feasibility through different safe sets; backup CBFs keep racing drones inside a geofence [8], and the braking-distance barrier of adaptive cruise control [9] is the special case of a constant authority. Cable-robot wrench calculations describe forces available at a fixed geometry, as intervals or zonotopes [10]–[12]; aerial transport models describe how that geometry moves [13], [14], and CBFs have been used to plan cable transport [15]. Concurrently, [16] certifies a quadrotor that must reorient its thrust before it can brake, with a body-rate model of first order; here the authority follows from the cable geometry, the swing is of second order, and the same calculation gives an outer bound. Closed-form backup control is also studied in [5]; our closed

form concerns the future wall excursion for this particular authority model.

We let each quadrotor reserve part of its thrust for the cable tension and part for swinging the cable; the braking authority that this guarantees is the support function of the zonotope of tension forces and depends on the cable directions alone (Sec. IV). When every cable swings to its limit, the payload covers at most the stopping distance D toward the wall, which is computed exactly (Sec. IV-B). The authority barrier is $H := h - D$: the states with $H \geq 0$ and admissible swing states form a controlled-invariant set, in which the braking maneuver satisfies every constraint of the filter of Sec. V; two further barriers bound the altitude of the payload during the maneuver. With upper bounds on the authority and on the swing acceleration, the same calculation gives an optimistic stopping distance D^{rel} , below which no state is viable (Theorem 9). Both bounds hold for every system whose braking authority is bounded by nondecreasing functions of coordinates that the inputs drive only through their second derivative (Sec. IV-A); the cable system is one instance.

The contributions are: (i) a condition under which an initially feasible HOCBF filter must lose feasibility before a cable reaches the braking half-space (Theorem 3); (ii) an explicit stopping distance for the input-constrained taut-cable model, the authority and altitude barriers built from it, and the certified set on which the barrier filter is always feasible (Theorem 9(i), Proposition 12); (iii) an explicit outer bound on the viability kernel (Theorem 9(ii)); and (iv) simulations that test the certificate on its own model and measure its conservatism and the failure of a direct

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sampled implementation. In simulation, the HOCBF filter lost feasibility in all 690 trials; on the exact model with a 1 ms hold the authority filter reached the wall in one of 100 trials started at the boundary of its certified set, by $2\text{ }\mu\text{m}$; on the full-order model with a 5 ms hold and the wall barrier alone it reached the wall in 59 of 100. The sampled results concern the implementation and give no safety guarantee.

II. PROBLEM FORMULATION

Quadrotor $i \in \{1, \dots, N\}$ has position $p_i \in \mathbb{R}^3$, mass m_i , and thrust vector $u_i := f_i R_i e_3$, where $f_i \geq 0$ is the thrust magnitude, $R_i \in SO(3)$ the attitude, and e_3 the vertical unit vector; an attitude controller produces the commanded thrust vectors (Remark 14). Each thrust vector lies in the ball $\mathcal{F}_i := \{u \in \mathbb{R}^3 : \|u\| \leq f_{\max,i}\}$.

The payload is a point mass m_L at $x_L \in \mathbb{R}^3$, attached to quadrotor i by a massless, inextensible cable of length l_i with unit direction $q_i \in \mathbb{S}^2$ from the payload to the quadrotor, $p_i = x_L + l_i q_i$. The cable exerts the force $T_i q_i$ on the payload and $-T_i q_i$ on the quadrotor, with $T_i \geq 0$ because a cable cannot push. With the gravitational acceleration g ,

$$m_L \ddot{x}_L = \sum_i T_i q_i - m_L g e_3, \quad (1)$$

$$m_i \ddot{p}_i = u_i - m_i g e_3 - T_i q_i. \quad (2)$$

The state is $x := (x_L, \dot{x}_L, q, \dot{q})$ with $q := (q_1, \dots, q_N)$. We assume that every cable is taut, with a tension of at least $T_{\min} > 0$; the filter of Sec. V enforces this bound, and the simulations model slack cables.

We decompose each thrust vector along the cable and perpendicular to it, $u_i = s_i q_i + u_i^\perp$ with $s_i := q_i^\top u_i$, $u_i^\perp := P_i u_i$, $P_i := I - q_i q_i^\top$, and we use the *specific force* of the payload, $a := \ddot{x}_L + g e_3$.

Lemma 1 (Structure). *Along any solution of (1)–(2) with taut cables: (a) $T_i = s_i - m_i q_i^\top a + m_i l_i \|\dot{q}_i\|^2$, so the tension depends on the inputs only through the parallel components $s = (s_1, \dots, s_N)$; (b) $\ddot{q}_i = \frac{1}{m_i l_i} u_i^\perp - \frac{1}{l_i} P_i a - \|\dot{q}_i\|^2 q_i$, so the cable direction depends on the inputs only through $u_i^\perp$ and a ; (c) $M_a(q) a = \sum_i s_i q_i + \sum_i m_i l_i \|\dot{q}_i\|^2 q_i$ with $M_a(q) := m_L I + \sum_i m_i q_i q_i^\top \succ 0$, so the inputs determine a , and by (a) the tensions, uniquely and affinely.*

Proof. Differentiate $p_i = x_L + l_i q_i$ twice and substitute (1)–(2). Since $q_i^\top \dot{q}_i = 0$ and $q_i^\top \ddot{q}_i = -\|\dot{q}_i\|^2$, the projection of (2) along q_i gives (a) and the projection perpendicular to q_i gives (b); substituting (a) into (1) gives (c). $\square$

In words, the parallel component sets the tension, the perpendicular component swings the cable, the two share the limit $\|s_i q_i + u_i^\perp\| \leq f_{\max,i}$, and the quadrotors interact only through the payload's specific force. For fixed $(q, \dot{q})$, (a) and (c) define an affine bijection between s and $T = (T_1, \dots, T_N)$, so we use the tensions as decision variables: given T , $a = \frac{1}{m_L} \sum_j T_j q_j$, and (a) returns $s_i(T)$.

A. Constraints, viability, and the problem

We consider a wall with horizontal unit normal n pointing into it, with the constraint

$$h(x_L) := d_0 - n^\top x_L \geq 0, \quad (3)$$

where h is the distance from the payload to the wall. We write $v := n^\top \dot{x}_L$ for the approach speed, $y := -n$ for the braking direction, and $b := y^\top \ddot{x}_L$ for the braking deceleration, so that $\dot{h} = -v$ and $\dot{v} = -b$, and we describe each cable direction by the *swing coordinates* $z_i := q_i^\top y$ and $w_i := q_i^\top r$, $r := e_3 \times y$. A cable with $z_i > 0$ lies in the *braking half-space*: it leans away from the wall and pulls the payload away from it; a cable with $z_i < 0$ pulls the payload toward the wall.

We impose the operational constraints

$$\mathcal{X}_{\text{op}} := \{x : |z_i| \leq \bar{z}, (w_i, \dot{w}_i) \in \mathcal{V}(\nu_w, \bar{w}), \|\dot{q}_i\| \leq \bar{\omega} \forall i\}, \quad (4)$$

with $\bar{z}^2 + \bar{w}^2 \leq \sin^2 \theta_q$, $\theta_q < \pi/2$, which keep every cable within θ_q of the vertical, reserve stopping room for the transverse swing, and bound its swing rate; $\mathcal{V}$ is defined in (10) below. With the altitude $\chi := e_3^\top x_L$ of the payload and a band $[\chi_{\min}, \chi_{\max}]$, the constraint set is $\mathcal{C} := \{x \in \mathcal{X}_{\text{op}} : h \geq 0, \chi_{\min} \leq \chi \leq \chi_{\max}\}$. The inputs must respect the thrust limits, the tension bound, and a bound $a_{\max}$ on the payload's specific force:

$$\mathcal{U}(x) := \{u : u_i \in \mathcal{F}_i, T_i(x, u) \geq T_{\min} \forall i, \|\sum_i T_i(x, u) q_i\| \leq m_L a_{\max}\}. \quad (5)$$

The last constraint is a design limit on the payload's acceleration; the bound that follows from the thrust limits alone is too large for Assumption 4 to hold with realistic constants.

A state is *viable* if some input signal with values in $\mathcal{U}(x)$ keeps the trajectory in $\mathcal{C}$ forever; the set of viable states is the *viability kernel* $\text{Viab}(\mathcal{C})$ [17], outside of which no controller can keep the system safe. A set is *controlled invariant* if some admissible input signal keeps every trajectory that starts in it inside it; every controlled-invariant subset of $\mathcal{C}$ lies in $\text{Viab}(\mathcal{C})$, which has no closed form. The problem is: (a) find a controlled-invariant set $S \subseteq \mathcal{C}$ whose membership can be tested online; (b) find an explicit set that contains $\text{Viab}(\mathcal{C})$; and (c) derive a safety filter that is feasible at every state of S .

III. LOSS OF FEASIBILITY OF HIGH-ORDER CBF FILTERS

For a unit vector y , the *exact directional authority* at x ,

$$\alpha_y^{\text{ex}}(x) := \max\{y^\top \ddot{x}_L(x, u) : u \in \mathcal{U}(x)\}, \quad (6)$$

is the largest acceleration that admissible inputs can give the payload along y ; the maximum exists because $\ddot{x}_L$ is affine in u and $\mathcal{U}(x)$ is compact and convex ($-\infty$ if empty). Since it is attained, for any function β of the state,

$$\{u \in \mathcal{U}(x) : b(x, u) \geq \beta(x)\} \neq \emptyset \iff \alpha_y^{\text{ex}}(x) \geq \beta(x). \quad (7)$$

Since h has relative degree two with respect to the tensions, HOCBFs [18], [19] apply the barrier condition twice. Let α_1 be a continuously differentiable class- $\mathcal{K}$ function, α_2 an extended class- $\mathcal{K}$ function, and $\psi_1 := \dot{h} + \alpha_1(h) = -v + \alpha_1(h)$. With $\dot{v} = -b$, the HOCBF condition $\dot{\psi}_1 \geq -\alpha_2(\psi_1)$ reads

$$b \geq \beta(x) := \alpha_1'(h) v - \alpha_2(\psi_1). \quad (8)$$

The HOCBF filter minimizes the change to the nominal command over $K_{\text{HO}}(x) := \{u \in \mathcal{U}(x) : b(x, u) \geq \beta(x)\}$ and, while $K_{\text{HO}} \neq \emptyset$, keeps the state in $\mathcal{C}_{\text{HO}} := \{h \geq 0, \psi_1 \geq 0\}$ [1]. By (7), $K_{\text{HO}}(x) \neq \emptyset$ if and only if the feasibility margin $\mu(x) := \alpha_y^{\text{ex}}(x) - \beta(x)$ is nonnegative.

Definition 2 (No-braking set). *The no-braking set for the braking direction y is $\Sigma_y := \{q \in (\mathbb{S}^2)^N : z_i \leq 0, i = 1, \dots, N\}$.*

In Σ_y no cable lies in the braking half-space, so no admissible input can decelerate the payload; the set has nonempty interior and contains the configurations of a team accelerating toward the wall.

Theorem 3 (Conditional loss of HOCBF feasibility). *Let $x_0 \in \mathcal{C}_{\text{HO}} \cap \mathcal{X}_{\text{op}}$ have $z_i(0) < 0$ for all i , $0 < v_0 < \alpha_1(h_0)$, and $\mu(x_0) > 0$. Let $\tau := \min_i(-z_i(0))/\bar{\omega}$ and $t_\psi := [h_0 - \alpha_1^{-1}(v_0)]/v_0$. If $t_\psi < \tau$, then every admissible trajectory that remains in $\mathcal{X}_{\text{op}}$ has a nonempty interval before τ with $h > 0$, $q \in \Sigma_y$, and $K_{\text{HO}} = \emptyset$. The property persists on an open neighborhood of x_0 if the input set varies continuously and stays nonempty there.*

Proof. Until τ , $z_i(t) \leq z_i(0) + \bar{\omega}t \leq 0$. Because $T_i \geq T_{\min}$, $b = m_L^{-1} \sum_i T_i z_i \leq 0$, hence $v(t) \geq v_0$ and $h(t) \leq h_0 - v_0 t$. The first time h reaches $\alpha_1^{-1}(v_0) > 0$ occurs by $t_\psi < \tau$. Immediately afterwards $\psi_1 < 0$ while $h > 0$. Since $\alpha'_1(h) \geq 0$ and α_2 is extended class- $\mathcal{K}$, $\mu = \alpha_y^{\text{ex}} - \alpha'_1(h)v + \alpha_2(\psi_1) < 0$. The inequalities are strict at the initial state and the authority maximum is continuous under the stated regularity, so they persist locally. $\square$

The initial-feasibility hypothesis cannot be removed. For example, as $\psi_1 \downarrow 0$ with $\alpha_1(h) = kh$ and $\alpha_2(s) = \epsilon s$, the required braking tends to $kv > 0$, whereas the available braking in Σ_y is nonpositive; such initial states already have an infeasible QP. Thus a universal claim for *every* choice of HOCBF gains would be too strong. For the two quadrotors of Fig. 1 with $m_L = m_i = l_i = 1$ in SI units, $T_{\min} = 1$ N, $\bar{\omega} = 2$ s⁻¹, $\alpha_1(h) = h$, $\alpha_2(s) = 20s$, $h_0 = 1$ m, and $v_0 = 0.85$ m/s, the floor tensions give $b = -1$ m/s², the HOCBF threshold is $\beta = -2.15$ m/s², $t_\psi = 0.176$ s, and $\tau = 0.25$ s; provided those tensions satisfy the thrust constraints, this is an initially feasible instance that loses feasibility.

IV. THE AUTHORITY BARRIER

To obtain a braking authority that depends only on the cable directions, we fix in advance how each quadrotor divides its thrust between tension and swinging. Throughout, $(z)_+ := \max\{0, z\}$ and $(z)_- := \max\{0, -z\}$.

Assumption 4 (Thrust reserve). *There exist constants $\bar{T}_i \geq T_{\min}$ (tension limit) and $\rho_i \geq 0$ (swing reserve) with $\sum_j \bar{T}_j \leq m_L a_{\max}$ such that, for every $x \in \mathcal{X}_{\text{op}}$, every T with $T_{\min} \leq T_j \leq \bar{T}_j$ for all j , and every $u_i^\perp \perp q_i$ with $\|u_i^\perp\| \leq \rho_i$, the thrust vector $s_i(T)q_i + u_i^\perp$ lies in $\mathcal{F}_i$.*

In words, each quadrotor can produce any tension up to $\bar{T}_i$ and, simultaneously, any perpendicular thrust up to ρ_i , and

such tensions respect the cone in (5); by Lemma 1(a), the assumption holds if $(\bar{T}_i + m_i a_{\max} + m_i l_i \bar{\omega}^2)^2 + \rho_i^2 \leq f_{\max, i}^2$.

Definition 5 (Directional authority). *Let the hover floors $T_i^h \in [T_{\min}, \bar{T}_i]$ satisfy $\cos \theta_q \sum_i T_i^h \geq m_L g$, and let $\phi_i(z) := \bar{T}_i(z)_+ - T_i^h(z)_-$. The directional authority of q along the braking direction y is*

$$\alpha_y(q) := \frac{1}{m_L} \sum_{i=1}^N \phi_i(z_i), \quad z_i = q_i^\top y. \quad (9)$$

A cable in the braking half-space contributes most at $\bar{T}_i$, a cable that leans toward the wall opposes least at its floor; each ϕ_i is nondecreasing, convex, and piecewise linear with one breakpoint. The floor is T_i^h and not $T_{\min}$ because tensions at the floors carry the payload for every cable configuration of $\mathcal{X}_{\text{op}}$: a cable that leans toward the wall and carries its share of the weight pushes the payload toward the wall, and the authority charges this push.

Lemma 6 (Authority as a support function). *The set $\mathcal{Z}(q) := \{\sum_i T_i q_i : T_i^h \leq T_i \leq \bar{T}_i\}$ of tension forces is a zonotope, and $m_L \alpha_y(q) = \max_{f \in \mathcal{Z}(q)} y^\top f$ is its support function along y . Under Assumption 4, every $b \in [-\alpha_{-y}(q), \alpha_y(q)]$ is produced by some input in $\mathcal{U}(x)$ at every $x \in \mathcal{X}_{\text{op}}$; hence $\alpha_y \leq \alpha_y^{\text{ex}}$.*

Proof. $\mathcal{Z}(q)$ is the Minkowski sum of the segments $[T_i^h, \bar{T}_i] q_i$, the support function of a Minkowski sum is the sum of the support functions, and $\max_{T \in [T_i^h, \bar{T}_i]} T z_i = \phi_i(z_i)$. Assumption 4 with $u_i^\perp = 0$ realizes every T in the box, and $b = \frac{1}{m_L} \sum_i T_i z_i$ is affine in T . $\square$

A. Systems with second-order authority

The construction uses two properties of the cable system, stated here for a general system with coordinates $\zeta_1, \dots, \zeta_N$: the braking authority is bounded above and below by non-decreasing functions of the coordinates, and each coordinate responds to the inputs only through its second derivative. For $\nu, \bar{\zeta} > 0$, the *admissible swing set*

$$\mathcal{V}(\nu, \bar{\zeta}) := \{(\zeta, \omega) \in \mathbb{R}^2 : -\bar{\zeta} + \frac{(\omega)^2}{2\nu} \leq \zeta \leq \bar{\zeta} - \frac{(\omega)^2}{2\nu}\} \quad (10)$$

is the convex set of $(\zeta, \dot{\zeta})$ from which a double integrator with $|\ddot{\zeta}| \leq \nu$ can keep $|\zeta| \leq \bar{\zeta}$ for all future time.

Definition 7 (Second-order authority). *Consider a system with admissible inputs, a constraint function h with $\dot{h} = -v$ and $\ddot{h} = -b$, a closed operational set $\mathcal{X}_{\text{op}}$, and twice continuously differentiable coordinates $\zeta_1, \dots, \zeta_N$ of the state. The system has second-order authority along h on $\mathcal{X}_{\text{op}}$ with data $(\underline{\alpha}, \bar{\alpha}, \nu, \nu^{\text{rel}}, \bar{\zeta})$ if:*

- (C1) $\underline{\alpha}$ is the sum of a constant and of N continuous, non-decreasing, piecewise-linear functions of the individual ζ_i ; $\bar{\alpha} \geq \underline{\alpha}$ is continuous and nondecreasing in each ζ_i ; and $b \leq \bar{\alpha}(\zeta)$ for every admissible input at every $x \in \mathcal{X}_{\text{op}}$.
- (C2) $|\dot{\zeta}_i| \leq \bar{\zeta}$ on $\mathcal{X}_{\text{op}}$, and $|\ddot{\zeta}_i| \leq \nu^{\text{rel}}$ for every admissible input at every $x \in \mathcal{X}_{\text{op}}$.

(C3) There is a map that assigns to every $x \in \mathcal{X}_{\text{op}}$ with $(\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu, \bar{\zeta})$ for all i , and to every choice of $\ddot{\zeta}_i^d \in [-\nu, \nu]$, an admissible input that produces $b = \underline{\alpha}(\zeta)$ and $\dot{\zeta}_i = \ddot{\zeta}_i^d$ for all i , and whose solutions remain in $\mathcal{X}_{\text{op}}$ as long as $(\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu, \bar{\zeta})$ for all i .

In words, no admissible input brakes harder than $\bar{\alpha}$ or swings faster than ν^{rel} , while a reserved part of the input swings every coordinate at any rate up to ν and brakes with $\underline{\alpha}$ at the same time. For such a system the wall constraint alone defines $\mathcal{C}_h := \{x \in \mathcal{X}_{\text{op}} : h \geq 0\}$.

B. The braking maneuver and its stopping distance

Fix a deceleration $\nu' \in [\nu/2, \nu]$. For $(\zeta, \omega) \in \mathcal{V}(\nu, \bar{\zeta})$, let $\tilde{\zeta}(t; \zeta, \omega)$ be the motion from (ζ, ω) to $(\bar{\zeta}, 0)$ that accelerates at ν and decelerates at ν' , held at $\bar{\zeta}$ afterwards: with $r := \sqrt{(2\nu\nu'(\bar{\zeta} - \zeta) + \nu'\omega^2)/(\nu + \nu')}$, $t_1 := (r - \omega)/\nu$, and $t_2 := t_1 + r/\nu'$,

$$\tilde{\zeta}(t) = \begin{cases} \zeta + \omega t + \frac{\nu}{2}t^2, & 0 \leq t \leq t_1, \\ \bar{\zeta} - \frac{\nu'}{2}(t_2 - t)^2, & t_1 \leq t \leq t_2, \\ \bar{\zeta}, & t \geq t_2, \end{cases} \quad (11)$$

if $\omega \leq 0$ or $\omega^2 < 2\nu'(\bar{\zeta} - \zeta)$; otherwise the motion decelerates from the start at the constant rate $\omega^2/(2(\bar{\zeta} - \zeta)) \in [\nu', \nu]$ and arrives at $t_2 = 2(\bar{\zeta} - \zeta)/\omega$. Because $\nu' < \nu$, the swing arrives through the interior of $\mathcal{V}(\nu, \bar{\zeta})$; the time-optimal swing, $\nu' = \nu$, is denoted $\tilde{\zeta}^\circ$. The swing is continuously differentiable in its initial data except on the curve $\omega^2 = 2\nu'(\bar{\zeta} - \zeta)$, $\omega \geq 0$, where its two cases meet and its derivatives are one-sided. A state is *regular* if none of its swing states lies on this curve, and *degenerate* otherwise.

Let x have $(\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu, \bar{\zeta})$ for all i , and let every coordinate follow its swing $\tilde{\zeta}_i(t) := \tilde{\zeta}(t; \zeta_i, \dot{\zeta}_i)$. Along the maneuver, the braking deceleration is $\bar{\alpha}(t; x) := \underline{\alpha}(\tilde{\zeta}_1(t), \dots, \tilde{\zeta}_N(t))$, the approach speed $\tilde{v}(t; x) := v - \int_0^t \bar{\alpha}(s; x) ds$, and the distance traveled toward the wall $E(t; x) := \int_0^t \tilde{v}(s; x) ds$. The *stopping distance* is

$$D(x) := \max_{t \geq 0} E(t; x), \quad (12)$$

with maximizing times $T^*(x)$; the maximum exists if $\underline{\alpha}_{\text{sat}} := \underline{\alpha}(\bar{\zeta}, \dots, \bar{\zeta}) > 0$, and it is taken over all $t \geq 0$, not up to the first zero of $\tilde{v}$, because a negative authority accelerates a payload at rest toward the wall. The *authority barrier* is $H(x) := h - D(x)$, and

$$\mathcal{X}_{\text{RF}} := \{x \in \mathcal{X}_{\text{op}} : H(x) \geq 0, (\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu, \bar{\zeta}) \forall i\}. \quad (13)$$

$H(x)$ is the smallest distance to the wall that the payload reaches during the maneuver started at x . The functions $h_i^+ := \bar{\zeta} - \zeta_i - (\dot{\zeta}_i^2)/(2\nu)$ and $h_i^- := \bar{\zeta} + \zeta_i - (\dot{\zeta}_i^2)/(2\nu)$ are nonnegative exactly on $\mathcal{V}(\nu, \bar{\zeta})$. Let $\pi^\#(x)$ be the input that (C3) assigns to x with $\ddot{\zeta}_i^d = \ddot{\zeta}(0; \zeta_i, \dot{\zeta}_i)$; since the swing is a feedback of $(\zeta_i, \dot{\zeta}_i)$, the closed loop under $\pi^\#$ reproduces the swings, $b(t) = \bar{\alpha}(t; x_0)$, and $v(t) = \tilde{v}(t; x_0)$.

Lemma 8 (Envelope and regularity). (i) Every $\zeta(\cdot)$ with absolutely continuous derivative, $\zeta(0) = \zeta$, $\dot{\zeta}(0) = \omega$, $|\dot{\zeta}| \leq \nu$

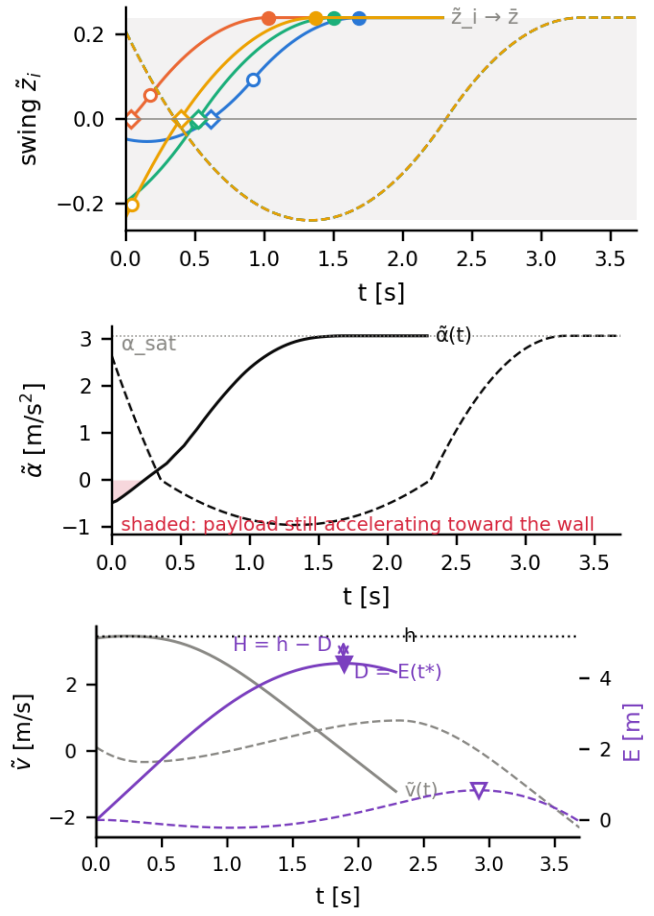


Fig. 2. A reserved swing (top) changes the cable directions and increases the available braking (middle). The barrier subtracts the maximum wallward excursion $\max_t E(t)$ from the current distance h (bottom). Solid and dashed curves are two example states.

a.e., and $\zeta \leq \bar{\zeta}$ satisfies $\zeta(t) \leq \tilde{\zeta}^\circ(t; \zeta, \omega)$ for all $t \geq 0$. (ii) D is locally Lipschitz on the states with $(\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu, \bar{\zeta})$ for all i , and along the maneuver started at x_0 , $D(x(t)) = \max_{s \geq t} \int_0^s \tilde{v}(\cdot; x_0) - \int_0^t \tilde{v}(\cdot; x_0)$.

Proof scheme. (i) Write $S(\zeta) := \sqrt{2\nu(\bar{\zeta} - \zeta)}$. A trajectory moving toward $\bar{\zeta}$ must satisfy $\dot{\zeta} \leq S(\zeta)$, or even full braking would overshoot. Before its switch, $\tilde{\zeta}^\circ$ uses the largest acceleration ν ; afterwards it follows $\dot{\zeta}^\circ = S(\bar{\zeta})$ to $\bar{\zeta}$. A first crossing from below either contradicts maximal acceleration before the switch or forces both motions to coincide on the stopping curve after it. (ii) On compact initial-state sets the arrival times and the maximizing times are uniformly bounded because $\underline{\alpha}_{\text{sat}} > 0$, and $E(t; \cdot)$ is locally Lipschitz uniformly in t ; the maximum of such a family is locally Lipschitz. Restarting the swing at time t reproduces its remaining trajectory, which gives the time-shift identity. $\square$

C. Main result

At regular states, for fixed t , $E(t; x)$ is continuously differentiable in $(v, \zeta, \dot{\zeta})$, because the integration over s smooths the breakpoints of $\underline{\alpha}$, which each $\tilde{\zeta}_i$ meets only at isolated times except while held at $\bar{\zeta}$. There, Danskin's

theorem [20] gives the directional derivative

$$D'(x; \delta) = \max_{t \in T^*(x)} \nabla_x E(t; x)^\top \delta \quad (14)$$

in directions δ along which the state stays in the domain of D . For an admissible u , let $F_u(x)$ be the state derivative and let $H'(x; F_u(x))$ be the upper right Dini derivative of H along it; at regular states it equals $-v - D'(x; F_u(x))$, the ordinary derivative $\dot{H}(x, u)$ when the maximizing time is unique. On states with all $(\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu^{\text{rel}}, \bar{\zeta})$, let D^{rel} be (12) computed with $(\bar{\alpha}, \nu^{\text{rel}})$ in place of $(\underline{\alpha}, \nu)$ and with the time-optimal swing: the stopping distance of an optimistic system that swings and brakes as hard as any admissible input allows. Let γ be a locally Lipschitz extended class- $\mathcal{K}$ function.

Theorem 9 (Inner and outer bounds). *Let the system have second-order authority along h on $\mathcal{X}_{\text{op}}$ with data $(\underline{\alpha}, \bar{\alpha}, \nu, \nu^{\text{rel}}, \bar{\zeta})$, and let $\underline{\alpha}_{\text{sat}} > 0$. Then:*

- (i) $\mathcal{X}_{\text{RF}}$ is controlled invariant. For every $x \in \mathcal{X}_{\text{RF}}$, the set $K_H(x)$ of the admissible u with $H'(x; F_u(x)) \geq -\gamma(H(x))$ and $\dot{h}_i^\pm \geq -c_V \sqrt{h_i^\pm}$ for all i , $c_V := 2\sqrt{2\nu\nu'}/(\nu - \nu')$, contains $\pi^\sharp(x)$. Every solution in $\mathcal{X}_{\text{op}}$ generated by a measurable selection $u(t) \in K_H(x(t))$ from $x(0) \in \mathcal{X}_{\text{RF}}$ satisfies $(\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu, \bar{\zeta})$ and $h(t) \geq H(x(t)) \geq 0$ for all $t \geq 0$.
- (ii) $\text{Viab}(\mathcal{C}_h) \subseteq \{x \in \mathcal{X}_{\text{op}} : (\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu^{\text{rel}}, \bar{\zeta}) \forall i, h \geq D^{\text{rel}}(x)\}$.

Proof. (i) Under $\pi^\sharp$ from $x_0 \in \mathcal{X}_{\text{RF}}$, Lemma 8(ii) and $h(t) = h_0 - \int_0^t v$ give $H(x(t)) = h_0 - \max_{s \geq t} \int_0^s v$, nondecreasing in t ; $\pi^\sharp$ keeps the state in $\mathcal{X}_{\text{op}}$ and the swing states in $\mathcal{V}$, so $\mathcal{X}_{\text{RF}}$ is forward invariant under $\pi^\sharp$, hence controlled invariant. Next, the swing restarted at a later state of the maneuver is the remaining part of the swing started at x , so the derivative of $E(t; \cdot)$ along $F_{\pi^\sharp}(x)$ equals $\tilde{v}(t; x) - v$ at regular states, and by (14), $H'(x; F_{\pi^\sharp}(x)) = -\max_{t \in T^*(x)} \tilde{v}(t; x)$. Every maximizing time $t > 0$ is a stationary point of $E(\cdot; x)$, so $\tilde{v}(t; x) = 0$, and $t = 0$ maximizes only if $v \leq 0$; hence $H'(x; F_{\pi^\sharp}(x)) \geq 0 \geq -\gamma(H(x))$ at regular states. The time-shift identity gives the same inequality for the Dini derivative along $\pi^\sharp$ at degenerate states. The swing of $\pi^\sharp$ satisfies the conditions on $h_i^\pm$: $\dot{h}_i^- \geq 0$; while it accelerates with $\dot{\zeta}_i > 0$, $\dot{h}_i^+ = -2\dot{\zeta}_i$ and $\dot{\zeta}_i^2 < 2\nu(\bar{\zeta} - \zeta_i) < 2\nu\nu' h_i^+ / (\nu - \nu')$; while it decelerates at a rate $a \in [\nu', \nu]$, $(\dot{h}_i^+)^2 = 2a(1 - a/\nu) h_i^+ \leq 2(\nu - \nu') h_i^+ \leq c_V^2 h_i^+$. Finally, along a solution generated by K_H , a function $h_i^\pm$ cannot become negative, since it would have a nonnegative derivative wherever it is negative; hence the swing states stay in $\mathcal{V}$, where D is locally Lipschitz, so $H(x(t))$ is absolutely continuous with $H'(x; \dot{x}) \geq -\gamma(H)$ a.e., and the comparison argument for nonsmooth barriers [21] gives $H \geq 0$; $h \geq H$ since $D \geq 0$.

(ii) Let $x_0 \in \text{Viab}(\mathcal{C}_h)$ and let an admissible input keep the solution in $\mathcal{C}_h$. By (C2), $(\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu^{\text{rel}}, \bar{\zeta})$ at all times, and by Lemma 8(i) with ν^{rel} , $\zeta_i(t) \leq \bar{\zeta}_i^{\text{rel}}(t)$. By (C1) and the monotonicity of $\bar{\alpha}$, $b(t) \leq \bar{\alpha}(\zeta(t)) \leq \bar{\alpha}(\bar{\zeta}^{\text{rel}}(t)) = \bar{\alpha}^{\text{rel}}(t; x_0)$, so $v(t) \geq \bar{v}^{\text{rel}}(t; x_0)$ and $\min_t h(t) \leq h_0 - D^{\text{rel}}(x_0)$; if $h_0 < D^{\text{rel}}(x_0)$, the solution leaves $\mathcal{C}_h$. $\square$

Together, (i) and (ii) give $\mathcal{X}_{\text{RF}} \subseteq \text{Viab}(\mathcal{C}_h) \subseteq \{h \geq D^{\text{rel}}\}$. The gap $D - D^{\text{rel}}$ is what the division of the input costs; for a constant authority $\underline{\alpha} \equiv \bar{\alpha} \equiv \bar{\alpha} > 0$ both distances equal $(v)_+^2 / (2\bar{\alpha})$, the braking-distance barrier of [9]. Part (i) is a backup-CBF guarantee [4], [22] with $\pi^\sharp$ as the backup controller, but the closest approach along the backup trajectory has the closed form (12) instead of being simulated, and the same data yield the outer bound (ii). The sets $\mathcal{X}_{\text{RF}}$ and $\mathcal{C}_{\text{HO}}$ need not be nested.

D. Application to the cable system

Assumption 10 (Swing reserve). *There exist $\nu, \nu_w > 0$ and $\rho_i^{\text{fb}} \geq 0$ such that, for every i , $\sec \theta_q \sqrt{\bar{\eta}_{z,i}^2 + \bar{\eta}_{w,i}^2} + \rho_i^{\text{fb}} \leq \rho_i$, where $\bar{\eta}_{z,i} := m_i(l_i \nu + a_{\text{max}} + l_i \bar{\omega}^2 \bar{z})$ and $\bar{\eta}_{w,i} := m_i(l_i \nu_w + a_{\text{max}} + l_i \bar{\omega}^2 \bar{w})$.*

Assumption 11 (Swing-rate consistency). $\bar{\omega}^2 \geq \sec^2 \theta_q (4\nu \bar{z} + 4\nu_w \bar{w})$.

Assumption 10 reserves, within ρ_i , the perpendicular thrust needed to impose any $\ddot{z}_i \in [-\nu, \nu]$ and $\ddot{w}_i \in [-\nu_w, \nu_w]$ against the coupling terms of Lemma 1(b), the factor $\sec \theta_q$ accounting for the non-orthogonality of the two swing directions on the sphere, and keeps ρ_i^{fb} for feedback; Assumption 11 makes the swing-rate bound of (4) hold on the admissible swing sets.

The maneuver also bounds the altitude. Its tensions lie in $[T_i^h, \bar{T}_i]$, so the vertical acceleration $\ddot{\chi} = \frac{1}{m_L} \sum_i T_i e_3^\top q_i - g$ lies between $c_0 := \frac{\cos \theta_q}{m_L} \sum_i T_i^h - g \geq 0$ and $c_1 := \frac{1}{m_L} \sum_i \bar{T}_i - g$ until the *stopping time* $t_s(x) := \max\{t_a + \bar{v}(t_a; x)/\underline{\alpha}_{\text{sat}}, t_a - \epsilon, 0\}$, $t_a := \max_i t_{2,i}$, with a lead $\epsilon > 0$ small enough that $z_i > 0$ whenever $t_{2,i} \leq \epsilon$; at t_s every swing is about to arrive and the payload recedes from the wall. From then on $\pi^\sharp$ holds the altitude with one uniform tension that decelerates a climb at $c_\downarrow \leq g - \frac{N}{m_L} T_{\min}$ and a descent at $c_\uparrow \leq \frac{\cos \theta_q}{m_L} \sum_i \bar{T}_i - g$. With $U := \dot{\chi} + c_1 t_s$ and $L := \dot{\chi} + c_0 t_s$, let $\Delta_\uparrow := (\dot{\chi} t_s + \frac{c_1}{2} t_s^2 + \frac{U^2}{2c_\downarrow})_+$ if $U > 0$ and $\Delta_\uparrow := 0$ otherwise; let $\Delta_\downarrow := 0$ if $\dot{\chi} \geq 0$, $\Delta_\downarrow := \dot{\chi}^2 / (2c_0)$ if $\dot{\chi} < 0 \leq L$, and $\Delta_\downarrow := -\dot{\chi} t_s - \frac{c_0}{2} t_s^2 + \frac{L^2}{2c_\uparrow}$ if $L < 0$. The *altitude barriers* are

$$H_\uparrow := \chi_{\text{max}} - \chi - \Delta_\uparrow, \quad H_\downarrow := \chi - \chi_{\text{min}} - \Delta_\downarrow,$$

and the *certified set* is $\mathcal{S} := \mathcal{X}_{\text{RF}} \cap \{H_\uparrow \geq 0, H_\downarrow \geq 0\}$.

Proposition 12 (The cable system has second-order authority). *Let Assumptions 4, 10, and 11 hold and let $\zeta_i := z_i$. Then (1)–(2) has second-order authority along h on $\mathcal{X}_{\text{op}}$ with $\underline{\alpha} = \alpha_y$ of (9),*

$$\bar{\alpha}(z) = \min \left\{ \frac{1}{m_L} \sum_i \phi_i^{\text{rel}}(z_i), \right. \\ \left. a_{\text{max}} \sec \theta_q \max_i (z_i)_+ \right\}, \quad (15)$$

where ϕ_i^{rel} is ϕ_i with $\bar{T}_i^{\text{rel}} := f_{\text{max},i} + m_i a_{\text{max}} + m_i l_i \bar{\omega}^2$; ν of Assumption 10; $\nu^{\text{rel}} := \max_i (\frac{f_{\text{max},i}}{m_i l_i} + \frac{a_{\text{max}}}{l_i} + \bar{\omega}^2)$; and $\bar{\zeta} = \bar{z}$. The map of (C3) sets $T_i = \bar{T}_i$ if $z_i > 0$ and $T_i = T_i^h$ otherwise, and steers each w_i within $\mathcal{V}(\nu_w, \bar{w})$.

Moreover, $\mathcal{S} \subseteq \mathcal{C}$ is controlled invariant, $\pi^\sharp$ satisfies $\dot{H}_\uparrow \geq 0$ and $\dot{H}_\downarrow \geq 0$, and Theorem 9(i) holds on $\mathcal{S}$ with these two conditions added to K_H .

Proof scheme. For (C1), $T_i \leq f_{\max,i} + m_i a_{\max} + m_i l_i \bar{\omega}^2$ follows from Lemma 1(a), and the cone of (5) gives $\sum_i T_i \cos \theta_q \leq e_3^\top \sum_i T_i q_i \leq m_L a_{\max}$, hence $b = \frac{1}{m_L} \sum_i T_i z_i \leq a_{\max} \sec \theta_q \max_i (z_i)_+$. Projecting Lemma 1(b) onto y establishes (C2). For (C3), choose the tensions that realize $\underline{\alpha}$ and prescribe $\ddot{z}_i \in [-\nu, \nu]$ plus a transverse $\ddot{w}_i \in [-\nu_w, \nu_w]$ that keeps w_i in its admissible set. The Gram matrix of $(P_i y, P_i r)$ has eigenvalues 1 and $1 - z_i^2 - w_i^2 \geq \cos^2 \theta_q$, so the required perpendicular thrust has norm at most $\sec \theta_q \sqrt{\bar{\eta}_{z,i}^2 + \bar{\eta}_{w,i}^2} \leq \rho_i$, and Assumption 4 realizes the tensions and swings simultaneously. Finally, $\|\dot{q}_i\|^2 \leq \sec^2 \theta_q (\dot{z}_i^2 + \dot{w}_i^2) \leq \bar{\omega}^2$ on the product of the admissible swing sets, so the trajectory stays in $\mathcal{X}_{\text{op}}$. For the altitude, $\ddot{\chi} \in [c_0, c_1]$ while $\pi^\sharp$ brakes; comparison with the constant accelerations c_1 and c_0 until t_s , and with the decelerations of the hold afterwards, bounds the climb by $\Delta_\uparrow$ and the drop by $\Delta_\downarrow$. Along $\pi^\sharp$, each of the two arguments of t_s decreases at unit rate while it is positive, so $\dot{H}_\uparrow = (t_s + U/c_\downarrow)(c_1 - \ddot{\chi}) \geq 0$ where $\Delta_\uparrow > 0$, and likewise for $H_\downarrow$; the hold keeps $b > 0$, so D is unchanged. $\square$

Hence $\mathcal{S} \subseteq \text{Viab}(\mathcal{C}) \subseteq \text{Viab}(\mathcal{C}_h) \subseteq \{h \geq D^{\text{rel}}\}$, which answers problems (a) and (b); Sec. V answers (c). With all cables at $\bar{z}$, $D^{\text{rel}} = (v)_+^2 / (2a_{\max} \bar{z} \sec \theta_q)$ against D = $(v)_+^2 / (2\alpha_{\text{sat}})$ with $\alpha_{\text{sat}} = \frac{1}{m_L} \sum_i \bar{T}_i \bar{z}$; where the cables must first swing, the gap is set by ν against ν^{rel} .

Proposition 13 (Computation). *For $t_j \in T^*(x)$, the time derivative of $h - E(t_j; \cdot)$ under u is*

$$\begin{aligned} \dot{H}_j(x, u) = & -v + t_j b \\ & - \sum_i (\partial_{z_i} E(t_j; x) \dot{z}_i + \partial_{\dot{z}_i} E(t_j; x) \ddot{z}_i), \end{aligned} \quad (16)$$

with $b = \frac{1}{m_L} \sum_i T_i z_i$ and $\ddot{z}_i = \frac{1}{m_i l_i} y^\top u_i^\perp - \frac{1}{l_i} y^\top P_i a - \|\dot{q}_i\|^2 z_i$, and $H'(x; F_u(x)) = \min_j \dot{H}_j(x, u)$ at regular states. D, T^* , and the coefficients of (16) depend on the state only through $(v, \{z_i, \dot{z}_i\}_i)$ and are computed exactly in $O(N \log N)$ arithmetic operations, because $E(\cdot; x)$ is piecewise quartic with at most $4N$ breakpoints; the gradient rows take a further $O(N|T^*|)$.

Proof scheme. At a regular state, $\partial_v E(t; x) = t$ and the chain rule gives (16); Danskin's formula makes the derivative of $H = h - \max_t E(t)$ the minimum over its active times. Every $\ddot{z}_i$ is piecewise quadratic, with at most its switch, arrival, and two zero crossings as breakpoints. Thus $\ddot{\alpha}$ is piecewise quadratic, E piecewise quartic, and candidate maximizers are interval endpoints or roots of the cubic $\ddot{v}$; sorting $O(N)$ events takes $O(N \log N)$ operations. $\square$

Remark 14 (Scope of the guarantee). The certificate assumes an exact taut-cable model, continuously updated thrust commands, a feasible transverse swing reserve, and the existence of closed-loop solutions. Wind, mass errors, actuator tracking,

slack cables, and zero-order-hold inputs require additional error bounds and are *not* covered by Theorem 9. In particular, continuity and local Lipschitzness of D do not by themselves give a numerical sampled-data margin.

V. SAFETY FILTER

The decision variables are the tensions $T \in \mathbb{R}^N$ and the perpendicular components $u^\perp = (u_1^\perp, \dots, u_N^\perp)$, $u_i^\perp \perp q_i$; the thrust vector of quadrotor i is $s_i(T)q_i + u_i^\perp$ with $a = \frac{1}{m_L} \sum_j T_j q_j$ in Lemma 1(a). For each swing coordinate $\zeta \in \{z_i, w_i\}$ with data $(\nu_\zeta, \bar{\zeta}_\zeta) \in \{(\nu, \bar{z}), (\nu_w, \bar{w})\}$, let $h_\zeta^\pm$ be the two functions of Sec. IV-B that describe $\mathcal{V}(\nu_\zeta, \bar{\zeta}_\zeta)$, and let $c_a \geq 1$. Given the nominal thrust vectors u_i^{nom} , the *authority filter* solves

$$\min_{T, u^\perp} \sum_i \|s_i(T)q_i + u_i^\perp - u_i^{\text{nom}}\|^2 \quad (17a)$$

$$\text{s.t. } u \in \mathcal{U}(x), \quad (17b)$$

$$\dot{H}_j(x, T, u^\perp) \geq -\gamma(H(x)) \quad \forall t_j \in T^*(x), \quad (17c)$$

$$\dot{h}_\zeta^\pm \geq -c_V \sqrt{h_\zeta^\pm}, \quad |\ddot{\zeta}| \leq c_a \nu_\zeta, \quad (17d)$$

$$\dot{H}_\uparrow \geq -\gamma(H_\uparrow), \quad \dot{H}_\downarrow \geq -\gamma(H_\downarrow), \quad (17e)$$

for every i and every swing coordinate. Constraint (17c) is the barrier condition of Theorem 9(i) written with Proposition 13, one affine constraint per maximizing time. The first condition of (17d) keeps the swing states in their admissible sets; it is affine in ζ , and the swing of $\pi^\sharp$ satisfies it because it decelerates at $\nu' < \nu$, whereas the time-optimal swing reaches the boundary of $\mathcal{V}$ in finite time and satisfies no such condition. The second condition is the range of swing accelerations that $\pi^\sharp$ uses; it limits the change of the swing rates within one sampling period (Sec. VI). Constraint (17e) holds the altitude barriers of Sec. IV-D; their derivatives are affine in $(T, u^\perp)$ through $\ddot{\chi}$ and the derivative of t_s . At regular states the program is a second-order cone program with $3N$ variables, coupled through the common payload force a , the thrust reconstructions $s_i(T)$, and the cone in (5).

By Theorem 9(i) and Proposition 12, $\pi^\sharp(x)$ satisfies every constraint at the regular states of $\mathcal{S}$, with slack at least $\gamma(H)$ in (17c); at degenerate states the gradient rows are not asserted to characterize the Dini condition, and the filter applies $\pi^\sharp$ instead. Every closed-loop solution that starts in $\mathcal{S}$ stays in it, so $h \geq H \geq 0$ and $\chi_{\min} \leq \chi \leq \chi_{\max}$. The sampled implementation replaces the first condition of (17d) by the requirement that every swing state lies in its set $\mathcal{V}$ at the next sample, which is a second-order cone constraint.

VI. NUMERICAL STUDY

Figure 2 illustrates the predicted swing profiles. The team has four 1.5 kg quadrotors carrying a 1 kg point payload on 1 m cables, with $T_{\min} = 1$ N, $f_{\max} = 44$ N, $\bar{T}_i = 3.2$ N, $\rho_i = 34$ N, $a_{\max} = 13$ m/s², $\theta_q = 20^\circ$, $\bar{z} = 0.239$, $\bar{w} = 0.171$, $\bar{\omega} = 1$ s⁻¹, and $\nu = \nu_w = 0.5$ s⁻². The thrust-reserve bound holds: $\sqrt{(\bar{T}_i + m_i a_{\max} + m_i l_i \bar{\omega}^2)^2 + \rho_i^2} = 41.73$ N $< f_{\max}$. The code and videos of the trials are available online.¹

¹https://github.com/Hadi-Hajieghrary/Authority_Barriers

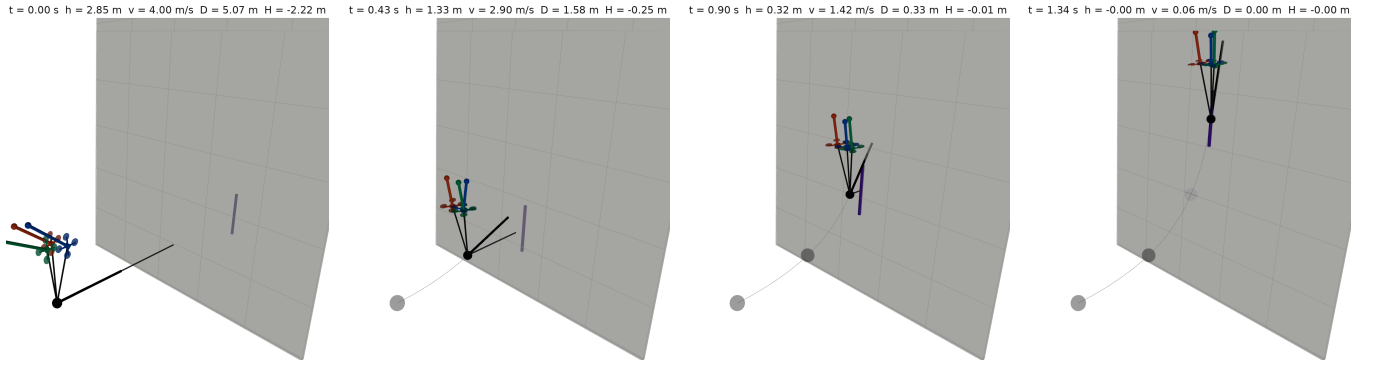


Fig. 3. Four frames of a collocation trajectory for the mixed cable configuration (B) at $v_0 = 4$ m/s from $h_0 = 0.56D$. The quadrotors reposition their cables behind the payload and enter $\mathcal{X}_{\text{RF}}$. The frames show one example of a safe trajectory; the kernel was not computed exhaustively.

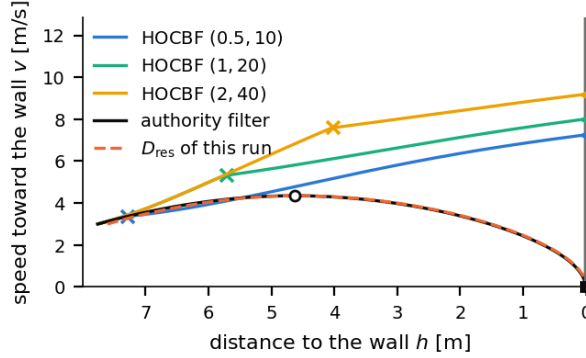


Fig. 4. Exact taut-cable model, one initial state of the certified set $\mathcal{S}$ and one adversarial nominal input, in the plane of wall distance and approach speed; the wall is the right edge. The HOCBF filters lose feasibility (crosses) while every cable leans toward the wall, and reach the wall (triangles). The authority filter brings every cable into the braking half-space (circle) and stops at the wall (square); the dashed curve shows D of its states.

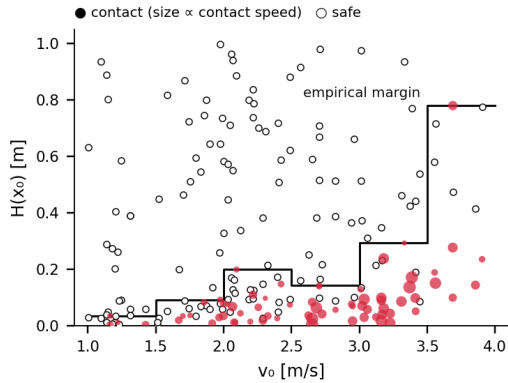


Fig. 5. Sampled-filter runs in the $(v_0, H(x_0))$ plane (filled: wall contact; open: no contact). The step gives the largest observed contact margin in each speed bin. The observed threshold is not a certified sampled-data bound.

Exact model. Theorem 9(i) was tested in its own setting: the authority filter (17) on the taut-cable model under a zero-order hold, with $T_i^h = 2.7$ N, $\nu' = 0.4$ s $^{-2}$, $c_{\downarrow} = 5$ and $c_{\uparrow} = 2$ m/s 2 , $\epsilon = 0.1$ s, $c_a = 1.25$, and the band $0 \leq \chi \leq 45$ m. The 100 initial states have random swing states, speeds of 1–4 m/s, and $H(x_0) \in [0, 0.05 D(x_0)]$, between 1 mm and 0.38 m; the nominal input is adversarial, the full thrust of every quadrotor toward the wall, tilted 40°

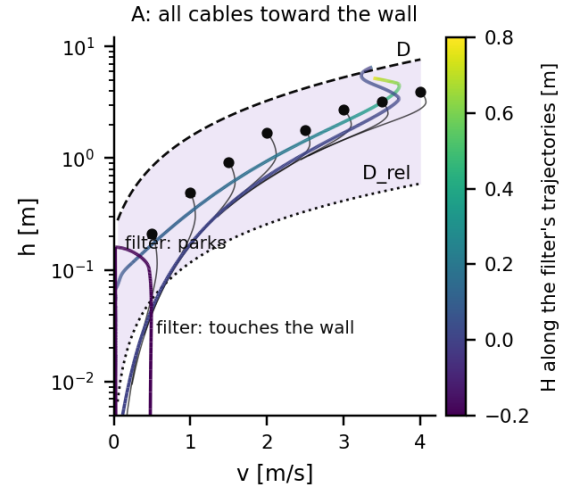


Fig. 6. Viability band for configuration A, cables initially leaning toward the wall: reserved (dashed) and optimistic (dotted) stopping distances. Dots are the smallest found and re-simulated collocation trajectories; they do not prove where the kernel boundary lies. Colored curves are two closed-loop trajectories of the sampled authority filter.

upward; a trial lasts 6 s. Table I gives the result. With a period of 1 ms the barrier stayed above -0.05 mm, one trial reached the wall, by $2 \mu\text{m}$, and the altitude stayed in its band in every trial; the payload climbed by up to 29.9 m. Figure 4 compares the authority filter with HOCBF filters with linear gains $(k_1, k_2) \in \{(0.5, 10), (1, 20), (2, 40)\}$ s $^{-1}$ from one state with $v = 3$ m/s, every cable at $z_i = -0.8\bar{z}$, and $h = 1.02 D = 7.76$ m. Each HOCBF program is feasible at the start and infeasible after 0.15, 0.52, and 0.78 s, with every cable leaning toward the wall; its relaxation with a penalized slack reaches the wall at 7.3–9.2 m/s. The authority filter has every cable in the braking half-space after 0.81 s and stops the payload 10 mm from the wall, with H between 0.15 and 0.01 m. Without the bound on the swing accelerations, the filter follows the nominal input with swing accelerations of about 20 s $^{-2}$, forty times ν ; the stopping distance then changes within one period by more than its gradient predicts, and the contacts multiply; a margin of the kind derived for sampled-data CBFs [23] is not available for D . With the hover floor the stopping distance from 3 m/s with every cable at $-\bar{z}$ is 9.18 m, against 6.56 m with the floor $T_{\min}$ and the

TABLE I

THE AUTHORITY FILTER ON THE EXACT TAUT-CABLE MODEL, 100 TRIALS
PER ROW.

Period	$ \ddot{\zeta} \leq c_a \nu_\zeta$	Contacts	Deepest	min H	χ in band
1 ms	yes	1	0.002 mm	−0.05 mm	100
5 ms	yes	4	0.4 mm	−7 mm	100
1 ms	no	6	0.1 mm	−13 mm	97
5 ms	no	33	1.7 mm	−104 mm	100

time-optimal swing: carrying the weight costs 2.6 m.

Full-order model. The remaining trials use rigid quadrotors, spring–damper cables that can go slack, and a filter at 200 Hz. They were run with the wall barrier alone, that is, with the floor $T_{\min}$ in ϕ_i , the time-optimal swing, and neither (17e) nor the bound on the swing accelerations; the altitude is free in them, and the payload climbs by 54 m in one trial before it stops. All 690 HOCBF trials lost QP feasibility before a cable entered the braking half-space: the three gain pairs of Fig. 4, each from 100 states with every cable leaning toward the wall at 2–4 m/s, with thrust vectors applied directly and with a 1 kHz attitude loop, and from 30 states with three quadrotors. This does not establish initial feasibility for arbitrary gains. Of 100 trials started with $H(x_0) \in [0, 0.05 D]$ under the adversarial nominal input, the sampled authority filter reached the wall in 59 despite a feasible QP at its sampling instants (Fig. 5); in 56 trials a cable was slack at some sample. The largest barrier loss was 0.85 m within one 5 ms hold; no contact was observed above an initial margin of 0.78 m in a second sweep of 100 states. This margin is empirical and holds for the tested distribution only. The attitude loop incurred 17–43 N tracking errors during command jumps.

Conservatism. Direct-collocation searches with three quadrotors covered three initial cable configurations at rest (A: $z_i = -0.15$; B: $z = (-0.15, 0, 0.15)$; C: $z_i = \bar{z}$) at eight speeds from 0.5 to 4 m/s, with each candidate subsequently simulated; like the full-order trials they constrain the wall distance only. When cables already lean away from the wall, the observed feasible distance is about 0.97–0.98 times D ; when a swing is required, feasible trajectories can begin at only 0.25–0.55 D (Fig. 6 shows configuration A). Figure 3 shows one of the found trajectories. The large gap is driven in part by the reserved swing acceleration: $\nu = 0.5 \text{ s}^{-2}$ needs 1.96 s to cross the allowed angle range, while the physical thrust can reach the swing-rate limit much sooner, and Assumption 11 admits no ν above 0.54 s^{-2} at this rate limit. A collocation success gives an example trajectory, but a failure does not locate the viability boundary, so the evidence supports the qualitative gap between the two models and validates neither set inclusion.

Cost. A backup filter that integrates $\pi^\#$ numerically [4] reproduced D within a relative error of $1.1 \cdot 10^{-6}$ and the commands of the filter within 0.4 % of $f_{\max}$ at 60 states, at 129 s per evaluation against 6 ms. In Python with the solver Clarabel, evaluating D takes 0.5–0.6 ms and the filter with the wall barrier 2–4.5 ms for two to eight quadrotors, close to the 5 ms sampling period; solver failures, model mismatch,

and inter-sample excursions preclude a real-time guarantee.

VII. CONCLUSION

Cable geometry makes braking authority a controlled state: the team brakes only as hard as the cable directions allow, and these change at a bounded rate. An initially feasible HOCBF filter can lose feasibility when the geometry cannot change quickly enough, under the stated gain-dependent condition. A support-function calculation and an explicit swing-and-brake maneuver give continuous-time inner and outer viability bounds, and the argument extends to every system with second-order authority. The maneuver carries the weight of the payload, which lengthens the stopping distance where cables lean toward the wall, and bounds the altitude. The inner bound is conservative where the cables must first swing: safe trajectories were found from as little as 0.25 D there, against 0.97–0.98 D with the cables already behind the payload; the reserved swing acceleration accounts for part of this gap.

On the exact model the certificate holds up to the hold: with a period of 1 ms and bounded swing accelerations, one of 100 trials started at the boundary of the certified set reached the wall, by 2 μm , and four did with 5 ms. The full-order trials, run with the wall barrier alone, a 5 ms hold, and cables that can go slack, do not satisfy the hypotheses of the theorem: the barrier lost up to 0.85 m within one hold, 56 of 100 trials had a slack cable, and 59 reached the wall. A certified sampled-data bound, full-order trials with the altitude barriers, a less conservative swing reserve, and a distributed implementation are the next requirements for a deployment claim.

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